

Task 2 Report Structure

Customer Retention & Churn Analysis

Data Science & Analytics Internship –
Future Interns

Submitted By:

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Task 2

Date:

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Objective

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The objective of this project is to analyze customer retention and churn patterns to identify customer behavior, retention trends, and factors affecting customer loss using data analytics and visualization techniques.

Dataset Information

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Dataset Name: Customer Churn Dataset

Number of Rows: 7000+

Number of Columns: 20+

important Columns:

- Customer ID
- Gender
- Tenure
- Monthly Charges
- Total Charges
- Contract Type
- Internet Service
- Churn

Tools & Technologies Used

Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- VS Code

Data Cleaning Process

Data Cleaning Process

- Removed missing values
- Checked duplicate records
- Converted data types
- Handled inconsistent values
- Verified churn column data

Customer Churn Analysis

Add:

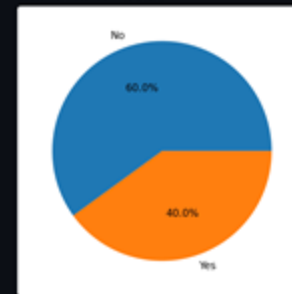
- Pie chart screenshot
- Churn distribution graph

Insights

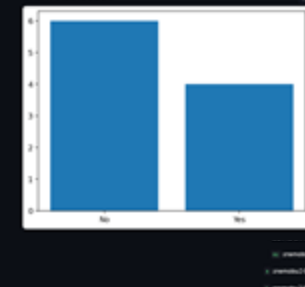
- A significant percentage of customers left the service
- Customers with month-to-month contracts showed higher churn
- Long-term contract customers had better retention

EXAMPLES

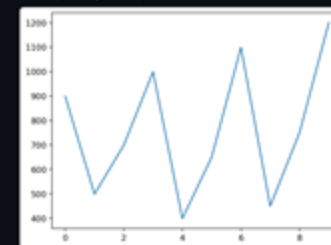
Customer Retention Pie Chart



Churn Distribution



Monthly Charges Analysis



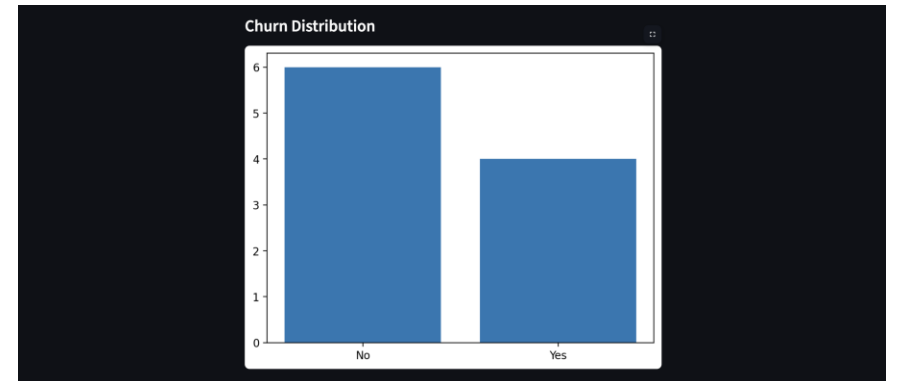
Contract Type Analysis

Add:

- Bar chart screenshot

Insights:

- Month-to-month contracts had the highest churn rate
- Two-year contracts showed strong customer retention
- Contract duration impacts customer loyalty



Monthly Charges Analysis

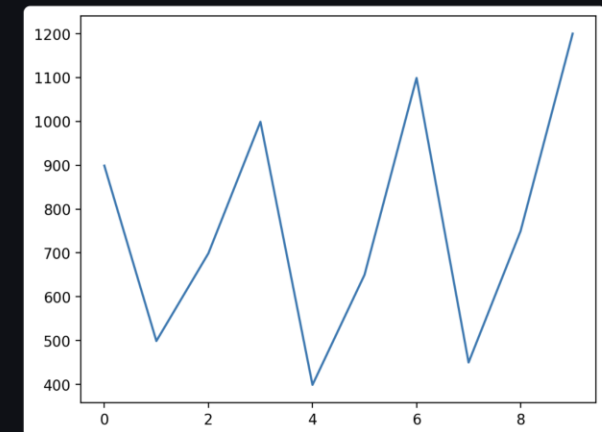
Add:

- Histogram/boxplot screenshot

Insights:

- Customers with higher monthly charges were more likely to churn
- Pricing influenced customer retention behavior

Monthly Charges Analysis



Recommendations

- Offer discounts for long-term contracts
- Improve customer support quality
- Introduce loyalty programs
- Analyze high-risk customers regularly
- Reduce churn through targeted retention campaigns

Conclusion

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This project helped identify customer churn patterns and key retention factors.

The analysis provided business insights to improve customer satisfaction

and reduce customer loss.

GitHub Repository

GitHub Repository:

https://github.com/dhathri-252/FUTURE_DS_02

Thank You

